

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457397.7687 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.046
Transit depth (Rp/Rs)^2	1.98 +/- 1.3 [%]
Orbital Inclination (inc)	78.19 +/- 2.95
Airmass coefficient 1 (a1)	1.42 +/- 0.27
Airmass coefficient 2 (a2)	-0.184 +/- 0.033
Scatter in the residuals of the lightcurve fit is	18.55 %
Best Comparison Star	None
Optimal Aperture	7.4
Optimal Annulus	13.57
Transit Duration (day)	0.022 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.046 vs 0.1368 (deviation 0.09σ)
Duration fit vs published	0.022 d vs 0.0758 d (deviation 1.86σ)
Mid-time O–C	4.5 min
Depth SNR	3.1
Residual scatter	18.55 %

Light curve

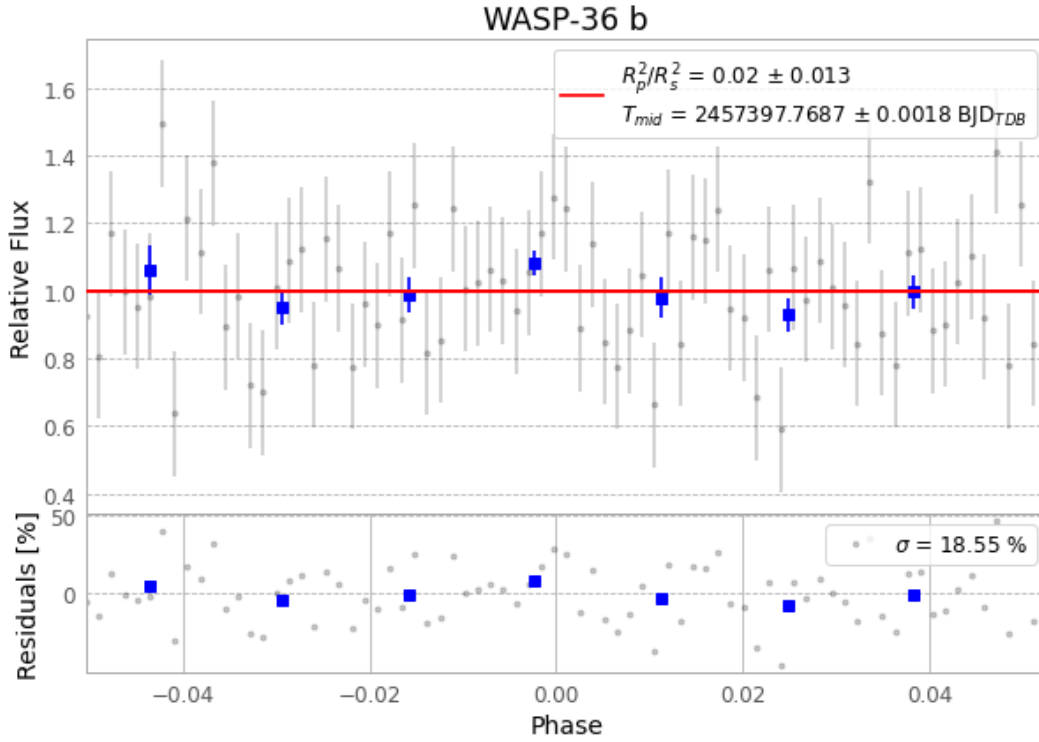


Figure 1 — FinalLightCurve_WASP-36 b_2016-01-09.png (SHA-256 e2de31eea979ee85...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [372, 367], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 6a1404caccd9307d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

77 FITS frames (51 MB), WASP-36160110042715.FITS → WASP-36160110081512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-160110.log (7a95d23cfcdf...), FinalLightCurve_WASP-36 b_2016-01-09.png (e2de31eea979...), FinalLightCurve_WASP-36 b_2016-01-09.csv (0b41ea2f07af...)

Compute resources

Wall time 102.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)